

Common Item 1 and 2 SCENARIO QUESTIONS
UCE MATHEMATICS 2024

Item 30

Achakara is a farmer who deals in commercial farming. She keeps animals in her farm for sale. Achakara has discovered that the prices of animals, goats and cows have gone up in the market. She plans to begin selling those two types of animals starting next week so that she can catch up with current market prices of the animals. She intends to sell at least 40 goats and at least 10 cows. She wants to sell not more than 160 animals altogether and also wants to ensure that she does not sell more cows than goats. In the market, the profit on the sale of each goat rose from UGX1800 to UGX3000 and the profit on the sale of each cow increased by UGX500 from UGX1500.

Task

- a) Write down four mathematical statements which represent these statements
- b) How many animals of each type should be sold by Achakara to make maximum profits? What is the maximum profit?

Item 31

In a certain village, there is a local shop that specializes in selling cooking oil. On a particular day, the shop attendant decided to set the price of frying oil at *shs. P* per litre. That day, he managed to sell L litres of oil, resulting in a total revenue of *shs. 8,000* from these sales. The shop was quite popular among the locals due to its competitive pricing and quality products. The following day, noticing an opportunity to attract more customers, the attendant reduced the price of frying oil by *shs. 50* per litre. This strategic decision paid off as it led to an increase in sales; specifically, he sold 4 more litres than the previous day. Consequently, on this second day of sales, his total revenue soared to *shs. 23,700*. It takes the shop attendant 30 minutes to cycle 10km when taking her product to the shop from home and an hour to cycle 15km. The shop is 35km away from home.

Task

- a) Determine the amount of oil sold before the price was reduced.
- b) Calculate the initial price per litre of the oil.
- c) How long does it take the shop attendant to cycle from home to the shop and back?

Item 32

A certain production unit of a manufacturing industry uses two types of machines, P and Q . Machine P requires $2m^2$ of floor space for installation, and 4 operators while machine Q requires $3m^3$ of floor space and 3 attendants. There are of $18m^2$ space and 24 attendants available. For an effective production, there should be more machines of type Q than those of type P . The report made by the marketing department in the industry shows that the profit generated by machine P is shs. 4,000 and that of machine Q is shs. 6,000.

Task

- Write down five mathematical statements from the information above.
- determine the number of machines of each type that has to be used in order to maximize the profits, state the maximum profit

Item 33

In a certain secondary school, there are two streams, A and B of senior three class students. During the opening weeks of the term, the class teacher counted the number of students and found that the ratio of the number of students in stream A to the number of students in stream B was 2:3 in the first week of the term. By the end of the second week when all the senior three students had reported, he counted the number of students again and discovered that 3 more students joined stream A. The class teacher thought of restreaming the students by distributing the students equally into the two streams. His target was to ensure that stream A has the same number of students as stream B. when the class teacher consulted the members of the class on why some students reported late to school, he was told that it was because some students didn't have school fees to start the term in the opening week. The class teacher was convinced because he knows that for every stream B student to be allowed in the school, he/she must have paid 0.4545454545 of the school fees and every stream A student must have paid 0.6666666 of the school fees. every stream A student must pay UGX2,200,000 before he/she is allowed to report to school in the opening week.

Task

- how many senior three students were in school in the first week?
- How many students should be in each stream so that they both have the same number of students?

- c) How much money do they allow every stream B student to pay as school fees before he/she is allowed in school in the opening week?

Item 34

The head teacher of a secondary school was so concerned about poor performance of a group of senior five students in end of term mathematics examinations. He therefore instructed all the mathematics teachers of the class to hold a brief meeting to find out the key reasons behind the poor results. This made the teachers to hold a meeting with the students in the main hall. The ratio of female to male students in the hall was 4:1. However, after some minutes, eight students left the hall and went to the school canteen to purchase a hot drink and snacks which made the ratio in the main hall to reduce to 8:1. The students left the main hall and walked $(\sqrt{5} - 2)$ metres west and $(\sqrt{5} + 2)$ metres north. The eight students later came back using the direct route and it took them 8minute to reach the main hall.

Task

- a) How many students were in the meeting?
b) Work out the speed that the eight students used to walk from the canteen back to main hall.

Item 35

A man called Joshua from Pukica village has only two sons, Walter and Frank. Johua owns a large rectangular plot of land of dimensions $2\sqrt{7}km$ by $2\sqrt{5}km$. He wants to distribute the plot of land equally among his two sons. There will be a fence that runs from one end of the diagonal of the plot of land to the other end so that it is equally distributed. Walter being the older son suggested that in some years to come, his father should buy for him more plots of land. When asked how hold he is, he stated that his age is a two-digit number whose sum of the ones digit and the tens digit is 10 and that if the digits are reversed, the number formed exceeds the original number by 54. Meanwhile, Frank is very young because his age is only one fourth of the brother's age.

Task

- a) How long is the fence that will be used to divide the plot of land?
b) Given that the man also wants to fence the boundary of the plot of land, determine the amount of fencing materials needed to achieve this.

c) Calculate the product of the ages of the two sons

Item 36

Joanilah owns a fruit store in her neighborhood. She sells fruits of two types; mangoes and pineapples. On the first day of last week, she sold mangoes and pineapples in the ratio of 2: 3. However due to customer care, she received more customers which made her to sell 12 more mangoes on the second day making the ratio of the number of mangoes to pineapples she sold in that week to be 10:9. She discovered that the amount of money collected from the sales of the fruits from the customers is partly constant and partly varies as the number of number of fruits sold after realizing that the cost of 50 fruits is *UGX*55000 and that of 80 fruits is *UGX*85000. When Joanilah asked one of her day to day customer, Monica on why she is always buying fruits, Monica explained that her family members love eating fruits a lot. She said that all the family members first eat all the three categories of fruits Mangoes, Pineapples and oranges at 6:30am and continue to eat mangoes, oranges and Pineapples at 8:00am, 10:30am and 12:00am respectively in a day. Joanilah got concerned on whether the family members get another time when they eat all the three categories of fruits again.

Task

- a) How much money did Joanilah collect from the fruits she sold last week?
- b) At what time do all the family member eat all the three fruits again at the same time

Item 37

Benjamin is planning to celebrate his birthday this coming month. He wants to ensure that everyone available at the party venue takes a cup of an orange drink that he will make. Benjamin uses 15ml of water to make a cup of an orange drink by mixing orange juice, water and honey in the ratio of 2:5:3. Each ml of water is sold at *UGX*20, each ml of orange juice at *UGX*150 and each ml of honey at *UGX*250. 588 people are expected to attend the celebration. The guests will be divided in to three groups; ordinary, VIP and VVIP to make the celebration special. All the guests will have drinks at the same time at exactly 8:00am and the ordinary group will continue to drink after every 40 minutes, the VIP after every hour and the VVIP after every 180minutes. The party is expected to end at 10:00pm so that everyone can return home happily.

Task

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- a) How much money will Benjamin incur in making the orange drinks?
- b) How many times will the three groups have drinks at the same time on that day?

Item 38

Bena is planning a head of the family gathering that will be held this weekend. She plans to go to the market to buy rice and beans for the gathering. Bena intends to buy at least 10kg of rice and beans altogether but she is not aware of the cost of a kg of beans and rice. After consulting her neighbor, she was told that the cost of rice is *shs 3000 per kg* while that of beans is *shs 2000 per kg* in the market. She intends to buy intends to buy at least more rice than beans and she has only *shs 30000* to spend.

Task

- a) Write down five mathematical statements to represent the given information.
- b) Show the feasible region of the statements on a Cartesian plane
- c) Find the maximum mass of each commodity Bena bought and how much money did she spend?

Item 39

Otema has two children, Oscar and Ruth who are sponsored by a certain non-governmental organization in school. The officials from the organization have discovered some missing details of the two children regarding their exact dates of birth and ages. They insist that Otema should provide them with the missing details since each child will soon be given a plot of land if he/she is found to be at least 8 years old but Otema only remembers that the difference squares of the ages of Oscar and Ruth is 65. When he picked a calculator to find more about the ages, he discovered that Oscar is 5 years older than Ruth. The plot of land is in form of a rectangle whose one side is 3m less than twice its width, and the area is $35m^2$. The plot of land is already fenced with colorful trees after every 3m from each other to stop other people from encroaching it.

Task

- a) Basing on calculations, find out whether or not the two children will be given the plot of land
- b) How many trees were used in fencing the plot of land?

Item 40

Ruth is done with her Advanced Level education. She has decided to start a small business of making cakes in her vacation to raise money for tuition fees in case she joins a University. She has decided that she is going to bake chocolate cakes and carrot cakes since they are on high demand in the market. She wants to bake at least 2 chocolate cakes but she wants to ensure that she bakes more carrot cakes than chocolate cakes. Due to limited time and other resources, she cannot bake more than 10 cakes. Ruth wants to sell each chocolate cake at UGX1500 and each carrot cake at UGX1000 in order to make sales of more than UGX12000.

Task

- a) Write down mathematical statements to represent the given information.
- b) Show the feasible region of the statements on a Cartesian plane
- c) Find the minimum number of cakes of each type Ruth can sell and still makes profits?

Item 41

Christopher recently bought a plot of land in Kampala. He made a decision to construct a small beautiful house on it to keep the land safe from encroachers. The construction required 35 tonnes and so, he hired a lorry and a truck that can carry 7 tonnes and 5 tonnes of sand each to the construction site. He paid UGX30000 per trip made by either a lorry or the truck to the construction site. Christopher had UGX180000 available for transportation. The trips made by the lorry did not exceed those made by the truck.

Task

- a) Write down mathematical statements to represent the given information.
- b) Show the feasible region of the statements on a Cartesian plane
- c) Find the number of trips that each of the vehicles made to ensure maximum amount of sand at the construction site?

Item 42

Jerome is a worker of a company in town. He loves saving money. Jerome saves money a fixed amount of money and continues saving more money at a rate that changes with the square of the number of days taken to save the money. It takes him 3 days to save UGX650,000 and 2 days to save UGX400,000. Jerome plans to save UGX36,000,00 and use it on the same day to buy a rectangular plot of land whose one

side is $5m$ more than twice the other side. The plot of land has an area of $250m^2$. Jerome wants to know how long one end of the diagonal of the plot of land is from the other end. He has decided that he will use a long rope to measure the length of the diagonals once he is done purchasing the plot of land.

Task

- a) How long will it take Jerome to save in order to meet her target?
- b) How long should that will be used in taking the measurement be?

Item 43

In the bustling new bus park located in Kampala city, the management has established specific parking regulations to optimize space and revenue. The park is designed to accommodate two types of vehicles: coasters and buses. Each coaster occupies $10m^2$ of space, while each bus requires $20m^2$. However, the area available for parking in the new bus park is limited to $500m^2$. To maintain order and efficiency, the park allows not more than 40 vehicles at any given time and at most 15 buses are allowed at a time. The daily parking charges are set at *shs* 5,000 for each coaster and *shs* 20,000 for each bus. The management wants to maximize their daily revenue from parking fees while adhering to all regulations.

Task

- a) Write down mathematical statements to represent the given information.
- b) Show the feasible region of the statements on a Cartesian plane
- c) Find the maximum daily revenue the management get from the parking of the vehicles and state the number of vehicles of each type that can be parked in park to achieve this.

Item 44

A group of boys of a secondary school was caught dodging lessons. The teacher on duty decided to give them work as a way of disciplining them. They boys were made to paint four sides walls of a building that measures $(\sqrt{7} - \sqrt{2})metres$ by $(\sqrt{7} + \sqrt{2})metres$. During a disciplinary hearing process, the teacher on duty was tasked by the deputy head teacher to provide report on the total number of students in school on that day. The teacher told the deputy head teacher that the number of boys to girls in the school was $(x + y) : (x - y) = 8 : 3$. To elaborate more about it, he said that there were 15 more boys than girls but the deputy head teacher insisted on knowing the

exact number of students available in school on that day but the teacher was unable to provide it.

Task

- a) Based on calculations, find out how much paint was used in painting the walls
- b) Provide the deputy head teacher with the exact number of students who were in school on the specific day

Item 45

Fredrick was invited to by his friend, Martin to go for a wedding ceremony in Adjumani district. He accepted the invitation but was confused on the means of transport to use since his car had a mechanical breakdown. Fredrick opted to hire a car and was told to pay for it in three successive weeks. According to the agreement, Fredrick will pay 8% of the agreed price in the first week, 0.324324324 of the price in the second week and the rest in the third week. He will pay UGX4,320,000 in the second week. The car had a registration plate number *UBC2331V*. Fredrick was told by Martin that the numerical digits in the number plate are in octal number system for security reasons and that the number plate that he should present to the police officers when told to must be in decimal system.

Task

- a) Calculate the amount of money Fredrick paid for the car in the first and third weeks
- b) Help Fredrick to identify the plate number that he should present to the police officers

Item 46

A local artist is creating a unique right angled triangular container to hold her handmade artworks. The dimensions of the container's sides are represented by the expressions $(2x - 3)m$, $(2x + 1)m$, and $(2x + 5)m$. The artist has noted that every $2.5m^2$ can hold 5 artworks when created. He plans to sell off all the artworks that the container can hold. 10 artworks are sold for UGX67000 in the local market and UGX15000 in the city market while 48 artworks are sold for UGX625000 in the city market and UGX314000 in the local market. The artist wants to sell all his artworks that the container can hold in the city market but will incur UGX85700 as transport cost.

Task

- a) Based on calculations, find how many artworks that the container can hold
- b) Calculate how much more money the artist will make from selling the artworks in the city market than in the local market.

Item 47

Samuel works with a factory than manufactures soft drinks. He has been complaining of too much fatigue due to too much time at his workplace since he works day and night. He has been saving UGX5200 every week to add on the UGX58000 that he already has in his bank account such that he can go and swim in a swimming pool in Naalya as a way of refreshing himself. The swimming pool in the shape of a trapezium with the two parallel sides being $3m$ and $5m$ respectively longer than the distance between them. The swimming pool holds a large amount of cold and clear water since it occupies an area of $32m^2$. The cost of swimming in the swimming pool for a day is UGX295,000.

Task

- a) Use an arrow diagram to show Samuel's saving pattern for the first 7 weeks and find how long it takes him to afford swimming in the swimming pool
- b) Calculate the perimeter of the swimming pool

Item 48

A secondary school in Uganda plans to allow its students to go back home for holiday. The term is about to end and the head teacher has tasked the school warden to ensure that all students reach their respective homes safely. The school warden has promised to take charge of the transportation of the students. He plans to use two vehicles for transporting back 800 students on the closing day of the term. He has a Costa of capacity 40 and taxi of capacity 20 students. The cost of each journey for a costa is shs. 20,000 and for the a taxi is shs15,000. The warden was told that the taxi cannot have twice journeys more than the costa. The warden does not have a lot of money for transportation. He has told the head teacher that the available amount for the transportation is 100,000/=.

Task

- a) Write down mathematical statements to represent the given information.
- b) Show the feasible region of the statements on a Cartesian plane

- c) Find the minimum amount of that can be spent on transport by the school warden.

Item 49

A house maid has noticed that in the previous weeks, water has been coming at a slow rate from the tap which made it take a long period of time to fill a tank. She has made up her mind to report this challenge to the owner of the family. She wants the owner of the family to have two taps A and B that fill a tank together in two hours and that the taps should be in such a way that tap A takes three hours less than B to fill the same tank. The maid is now interested in determining how long each tap can take to fill the tank before going to the owner of the family to render her request. She also needs to provide the owner of the family with the price of each tap and the total cost of the two taps. According to the maid the taps are not very expensive as 40 taps costs UGX205000 and 30 taps costs UGX155000.

Task

- a) How long does it take each of the two taps to fill the tank?
- b) Work out the cost of the two taps

Item 50

Mr. Magembe is a farmer who is always informed and updated about prices of various animals in the market. He has noticed that the prices of goats and rabbits are increasing at a fast rate as 20 goats now costs UGX5,185,800 and 25 goats now costs UGX6,475,800. He now wants to start constructing a farm in form of a rectangle for rearing goats. He wants to ensure that the farm is big enough and occupies an area of $750m^2$. Mr. Magembe is so scared of wild animals like leopards and lions attacking his goats on the farm. Because of this, he wants to ensure that the farm is well fenced. It will 110m of fencing materials to be used in fencing the farm so that the farm is safe from the wild animals. Very happy with his plan of constructing the farm, Mr. Magembe has discovered that the farm will not be able to rear very many goats as $8m^2$ of the farm can be used to rear only 2 goats.

Task

- a) Based on calculations, deduce the price of a goat and a rabbit in the market
- b) Work out the amount of money Mr.Magembe can earn if he decides to sell all the goats in the farm.